CA-WNN: Improving Weightless Neural Networks using Cellular Automata Feature Extraction

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***Abstract*— Cellular Automata (CA) and Weightless Neural Networks (WNN) are lightweight computational paradigms noted for their inherent parallelism and low computational complexity. This paper presents a hybrid image classification framework that integrates CA-based rule-driven feature extraction with a WNN classifier. The approach employs predefined morphological and edge detection rules to transform grayscale images into structured binary representations that capture spatial information. These features are encoded and stored in RAM-based WNN modules for efficient classification. The framework is evaluated on the MNIST dataset across multiple configurations, including variations in CA rules, iteration counts, and thermometer encoding levels. Sensitivity analysis is performed by varying the address size and pseudo-random mapping seeds to assess robustness. Experimental results show that the proposed CA–WNN model achieves a maximum test accuracy of 95.36%, outperforming baseline WiSARD and classical models such as Logistic Regression and SVM. Optimal performance is achieved with minimal CA iterations, ensuring efficiency and improved generalization.**

Keywords—Weightless Neural Network, Cellular Automata, Image Processing, Rule-based intelligence, Feature extraction.

I. Introduction

Cellular Automata (CA) were introduced by Von Neumann and Ulam, who represented a self-reproducing organism in 1946 [1]. Later, Stephen Wolfram expanded this concept by proposing elementary CA, which evolves based on local rules, to produce complex patterns from simple or random initial states [2]. John Conway proposed the widely recognized two-dimensional CA rule known as the Game of Life in 1970 [3]. With advancements in image processing, CA is used in image restoration, enhancement, feature extraction, edge detection, etc. [4, 5, 6]. Beyond visual tasks, CA are also widely used for simulations to model the spread of infectious diseases [7], wildfire propagation [8], urban land growth [11], etc., owing to their ability to model complex systems efficiently. Recently, CA was integrated into deep learning frameworks, utilizing its abilities of feature extraction [10, 15], pattern recognition [12, 13, 14], and classification [9, 15]. An increasing number of studies are investigating Convolutional Neural Network (CNN) based CA architecture for classification [16, 17, 18]. Replacing convolutional layers with CA is an active area of research to address the high data and computational demands of CNNs [19].

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Though Deep Neural Networks (DNN) achieved high accuracy in image classification, they require a significant number of weights and multiply-accumulate operations, causing high memory as well as energy requirements [20]. The Weightless Neural Networks (WNNs) that emerged between the 1950s and 1970s are a distinct alternative to conventional neural network models. Unlike conventional neural networks that modify the synaptic weights via backpropagation and gradient descent, WNN directly encodes and accesses patterns using a Random Access Memory (RAM) structure [22]. The artificial neuron in WNN is called RAM nodes that use lookup tables (LUTs) to represent a Boolean function of their inputs as a truth table. This methodology allows instant learning where new information is directly written into a memory location determined by input patterns, without the need for iterative or expensive floating-point computation. One of the early models of WNN was the Wilkie-Stonham-Aleksander Recognition Device (WiSARD), which was developed and introduced in 1981 [23]. Thermometer encoding, a unary representation technique, has been shown to improve classification performance in WNNs by preserving ordinal relationships in input data [24]. Recent works have demonstrated that WNNs can outperform traditional deep neural networks in terms of energy efficiency and latency, making them a compelling choice for deployment on edge devices [25, 26].

This paper presents a hybrid architecture combining CA for feature extraction with a WNN for classification. CA operates using predefined rules, incorporating morphological processes such as dilation and erosion, along with either traditional edge detection [27] or the Canny filter [28]. The proposed approach differs fundamentally from prior CA-integrated neural models in two respects. First, whereas recent works [16, 17, 18, 19, 21] embed CA as a learnable or differentiable component within a deep learning pipeline — retaining gradient-based optimization — the proposed method treats CA exclusively as a deterministic, fixed preprocessing stage. No CA parameters are learned; the rules are fully specified and interpretable by design. Second, the downstream classifier is a RAM-based WiSARD, which likewise requires no weight matrices, no backpropagation, and no floating-point multiply-accumulate operations. The result is an end-to-end training pipeline that is entirely free of gradient-based optimization, distinguishing it from all prior CA–DNN and CA–CNN hybrid architectures. This enables the extraction of complex spatial features through simple local interactions,

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which are subsequently mapped to a memory-based classifier, making the full framework well-suited for resource-constrained and interpretable edge applications. The effectiveness of the proposed framework is demonstrated on the MNIST dataset, where CA-enhanced features improve classification performance over a baseline WiSARD. The study also investigates the impact of key parameters, including the number of CA iterations, thermometer encoding levels, and discriminator address sizes, and compares the approach with SVM and k-NN classifiers.

The paper is organized as follows: Section 2 reviews CA and WNNs. Section 3 details the methodology, Section 4 discusses results, and Section 5 concludes.

II. Background

A. Cellular Automata

Cellular Automata are mathematical models used to study complex systems through simple, predefined rules. They consist of a grid of cells, each having a binary state, either 0 (dead) or 1 (alive). The state of each cell evolves over discrete time steps (generations) based on its current value and the values of its neighboring cells. In a one-dimensional CA, a cell's neighborhood typically includes its immediate left and right cells, and the next state is computed iteratively using the defined rule. Fig. 1 presents the two commonly used neighborhood definitions in a two-dimensional CA. The Von Neumann neighborhood includes a cell and its four immediate neighbors (up, down, left, right). The Moore neighborhood extends this by also including the four diagonal neighbors, resulting in a total of eight surrounding cells.

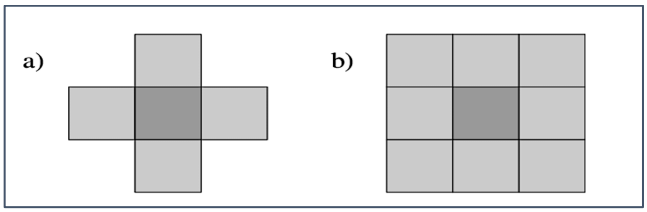


Fig. 1. The a) Von Neumann and b) Moore neighborhood in a 2-D CA.

CA offers several advantages over CNNs. Their local, inherently parallel computations yield greater efficiency for specific tasks and hardware implementations, often reducing model complexity while maintaining comparable performance with fewer parameters, smaller memory footprints, and faster training. The rule-based update mechanism also enhances interpretability, as decision processes can be directly traced to well-defined rules.

Nevertheless, CA presents notable limitations. They may underperform CNNs on tasks requiring deep feature abstraction from high-dimensional data, and their performance is highly sensitive to rule design, for which identifying optimal configurations remains an open challenge.

B. Weightless Neural Network

Weightless neural networks consist of RAM-based discriminators built from numerous one-bit memory units; each connected to the input through a predefined pseudo-random mapping. During training, memory locations corresponding to observed input patterns are activated, creating a stored representation of recognized patterns. When a novel input is presented, the discriminator retrieves and aggregates the relevant memory outputs, with higher cumulative values indicating a closer match to the training data. The WiSARD model comprises multiple RAM-discriminators, each trained to recognize a specific class. For a given input, each discriminator generates a response, which is evaluated using a confidence measure to determine the certainty of the classification.

Fig. 2 shows the diagram of a WiSARD model, initial input data is transformed into binary form and partitioned into a tuple of a predetermined size. These tuples serve as an address to retrieve data from RAM nodes. Each RAM functions as a discriminator that retains patterns it has encountered during training. During the classification, the network counts the number of active addresses that correspond to stored values, and the class with the highest count of matches is chosen as a prediction.

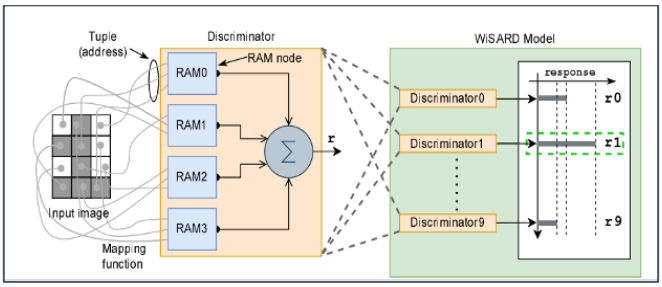


Fig. 2. RAM discriminator and WiSARD Model for classification, where the input image is pseudo-randomly mapped into the discriminator.

WNN offers several advantages over CNN in terms of computational efficiency and interpretability. WNNs are lightweight, requiring no weight matrices or gradient-based optimization, allowing faster training and low-power execution – ideal for resource-constrained environments. WNN can generalize from a smaller dataset, unlike CNN, which requires a large dataset. These are hardware-friendly due to simple logic and memory operations. The limitations are that WNN cannot learn hierarchical or abstract features. WNNs' reliance on binary input requires input conversion, losing meaningful information about the input.

III. Methodology

The proposed framework follows a sequential processing pipeline consisting of three primary stages: data binarization, CA–based feature extraction, and WNN classification.

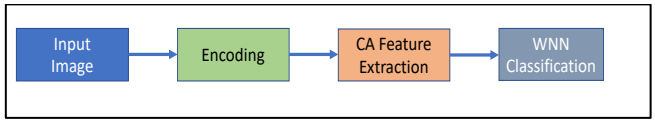


Fig. 3. Dataflow of the proposed hybrid CA–WNN architecture. Input images are first binarized using thermometer encoding. Three CA feature extraction models are evaluated: (a) the Game of Life automaton (GoL), (b) morphological CA rules including dilation, erosion, and edge detection, and (c) morphological CA rules with Canny edge detection replacing standard edge detection. Extracted features from each model are subsequently classified using the weightless neural network, enabling a comparative analysis of CA-based feature extraction strategies.

As illustrated in Fig. 3, the input images are first transformed through thermometer encoding, after which discriminative features are derived using CA dynamics. This combination enhances structural preservation and highlights local spatial patterns that are often overlooked by direct binarization alone. Finally, the extracted CA features are fed into a WiSARD-style WNN classifier, where each

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discriminator maps feature subsets to RAM nodes via pseudo-random addresses and performs class-specific pattern recognition, enabling efficient and accurate classification without relying on traditional weight-based learning.

A. Binarization: Thermometer Encoding

WNN requires binary inputs, so grayscale data must be converted without losing important information. Instead of simple thresholding, Thermometer Encoding is used. This method converts each pixel into multiple binary levels using increasing thresholds.

For a grayscale pixel $x_{i,j} \in [0, 255]$, the thermometer encoding at level $k \in \{1,2,\ldots,L\}$

$$b_{i,j}^{(k)} = \begin{cases} 1, & x_{i,j} \geq \tau_k \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where τ_k is the threshold for level k, typically defined as

$$\tau_k = \frac{255}{L+1} \cdot k \quad (2)$$

For instance, with $L = 4$ levels, each pixel is compared against four increasing thresholds, producing a set of binary images. Each binary plane indicates whether the pixel intensity exceeds a specific threshold, thereby preserving fine-grained intensity information across multiple levels and enhancing feature representation for subsequent CA-based processing.

B. Feature Extractor: Cellular Automata

The feature extractor processes each thermometer-encoded binary level independently, where each cell is represented in a binary state (1 for active/alive and 0 for inactive/dead), and applies predefined CA rules to enhance structural and spatial patterns. We investigate three distinct CA models:

1) Game of Life.

The well-known Conway's Game of Life automaton operates on a grid of cells, each in a binary state (alive or dead). The state of each cell in the next generation is determined by its eight neighboring cells according to the following rules:

a) Birth: A dead cell with exactly three live neighbors becomes alive.

b) Survival: A live cell with two or three live neighbors remains alive.

c) Death: A live cell with fewer than two (underpopulation) or more than three live neighbors (overpopulation) dies.

2) Morphological CA with Edge Detection.

Custom CA rules are applied for feature extraction, including:

a) Erosion Rule. A cell remains active only if all eight neighbors are active, effectively shrinking dense regions.

b) Dilation Rule. A cell becomes active if it or any of its eight neighbors is active, expanding active regions and filling small gaps.

c) Edge Detection Rule. A cell remains active if it was initially active and has some—but not all—neighbors active, highlighting boundaries of active regions.

Let $x_{i,j}^t \in \{0,1\}$ denote the state of a cell at iteration t, and let $x_{i,j}^{t+1}$ denote its next state. Let $S_{i,j}^t$ denote the number of active neighbors in the Moore neighborhood. The CA update rules are applied synchronously for each iteration.

The neighborhood sum is defined as

$$S_{i,j}^t = \sum_{(p,q)\in\mathcal{N}} x_{i+p,j+q}^t \quad (3)$$

where $\mathcal{N}$ is the 8-connected Moore neighborhood, p and q defines the row and column offset with respect to the central cell (i, j), excluding (0, 0).

The morphological rules are defined as:

Erosion

$$x_{i,j}^{t+1} = \bigwedge_{(p,q)\in\mathcal{N}} x_{i+p,j+q}^t \quad (4)$$

Dilation

$$x_{i,j}^{t+1} = \bigvee_{(p,q)\in\mathcal{N}} x_{i+p,j+q}^t \quad (5)$$

Edge Detection

$$x_{i,j}^{t+1} = x_{i,j}^t \wedge (0 < S_{i,j}^t < 8) \quad (6)$$

3) Morphological CA with Canny Edge Detection. This variant replaces the CA-based edge detection rule with the classical Canny edge detector, allowing a comparison between CA-driven and gradient-based edge extraction.

The use of these CA-based feature extraction strategies allows the system to capture both local structural patterns and global spatial information, enhancing the representation available to the WiSARD classifier. Fig. 4 shows features extracted using CA rules on a sample image of the digit '1'. The CA-WNN Model applies erosion, dilation, and edge detection to the raw input. Erosion thins the structures within the image, dilation thickens them, and edge detection captures the boundary.

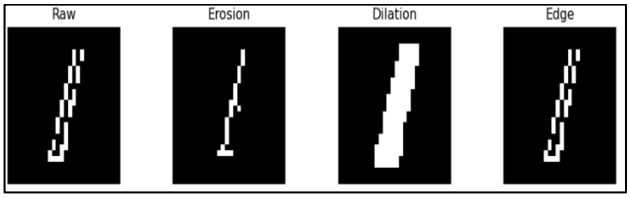


Fig. 4. Visualization of features obtained for CA-WNN using erosion, dilation, and edge detection.

For each thermometer-encoded level of the input image, each of these CA rules is applied sequentially. The resulting grids from all rules and all levels are then flattened and concatenated into a single, comprehensive binary feature vector and is fed into the classifier for final prediction.

C. Classification: Weightless Neural Network

The WiSARD architecture contains RAM neurons and multiple discriminators, with one discriminator for each class it needs to classify. Each discriminator consists of RAM neurons that serve as a lookup table.

The input to the network is a single, long binary feature vector. This feature vector is then divided into smaller, random subsets called address tuples. The number of features

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in each subset is determined by the address size, which is specified during initialization. For instance, a 9408-bit feature vector might be randomly split into 294 subsets, each with an address size of 32 bits. Each of these address tuples is then sent to a dedicated RAM neuron within a discriminator. The binary pattern of the tuple serves as a memory address, allowing the RAM neuron to access the corresponding memory location.

During the training phase, only the discriminator for a specific class (for example, the digit "5") is trained when a training image for that class is presented. For each RAM neuron in that discriminator, the addressed memory location is activated, and its value is either incremented or set to 1. This process essentially mimics one-shot learning, where the network "remembers" the specific patterns it has encountered.

In the classification phase, when a new, unseen image is presented, all discriminators receive the same input feature vector. Each discriminator calculates its response or activation by presenting the random address tuples to its respective RAM neurons and summing up the values of all the addressed memory locations. The discriminator with the highest response is then chosen as the predicted class. The higher the response, the more the input pattern resembles the patterns it was trained to recognize.

The core idea behind this approach is that a pattern belonging to a certain class will activate more of the RAM neurons in its corresponding discriminator compared to any other discriminator. The use of random mapping and multiple RAMs provides a form of generalization and noise tolerance, enabling the network to recognize patterns even if they differ slightly from the training examples.

IV. Experiments And Results

A. Dataset

The MNIST [29] dataset consists of 70,000 grayscale images of handwritten digits (0–9). Each image has a resolution of 28×28 pixels. The dataset is divided into 60,000 training samples and 10,000 testing samples.

B. Experiments and Results

The grayscale images are converted into a multi-binary representation using Thermometer Encoding. For each level, a pixel is set to 1 if its intensity exceeds the corresponding threshold, producing a stack of binary images that preserve fine-grained intensity variations.

Each of these binary images is then processed by a set of CA rules. Each rule produces a new binary image, and then the resulting feature maps are concatenated. The feature size can be calculated by multiplying the dimensions of the original image, the number of thermometer encoding levels, and the number of CA rules. It provides a dense feature vector that captures a rich set of structural and intensity-based patterns.

The concatenated feature vector is used as input to a WiSARD-based classifier with 10 discriminators, corresponding to 10 classes of the dataset. Each discriminator comprises multiple RAM nodes with a fixed address size of 32 bits. During training, each discriminator is updated exclusively with samples from its corresponding class. Input patterns are mapped to memory addresses using a pseudo-random mapping, and the addressed RAM locations are set to 1. During inference, the same mapping is applied across all discriminators. Each computes a score by summing the retrieved RAM values, and the class associated with the highest score is selected.

To account for the randomness in mapping, each experiment is repeated 5 times with different seeds, and the mean and standard deviation of accuracy are reported, as shown in Table 1. To assess the performance of a non-CNN architecture, a multinomial Logistic Regression model was employed with 200 iterations.

TABLE I. Performance Comparison of Different Models on the MNIST Dataset

Method	Feature	Train Accuracy	Test Accuracy
Neural Network	Logistic Regression	93.18 ± 0.0015	91.86 ± 0.0009
WNN (Baseline)	WiSARD without CA feature extraction	99.47 ±0.0003	91.16 ±0.0021
CA-WNN Mode 1	Game of Life	98.27 ±0.0061	86.36 ±0.0131
CA-WNN Mode 2	Morphological CA with edge detection (iteration = 2, level encoding = 4)	99.899 ±0.03	93.55 ±0.17
CA-WNN Mode 3	Morphological CA with edge detection (iteration = 2, level encoding = 8)	99.99 ±0.01	94.01 ±0.04
CA-WNN Mode 4	Morphological CA with Canny edge detection	95.36 ±0.03	93.56 ±0.04
CA-WNN Mode 5	Morphological CA with edge detection (iteration = 1, level encoding = 4)	99.98 ±0.01	95.04 ±0.01
CA-WNN Mode 6	Morphological CA with edge detection (iteration = 1, level encoding = 8)	100 ±0.00	95.36 ±0.04
CA-WNN mode 6 with Gaussian noise	Gaussian noise of mean 0 and std of 5 (iteration=1, level encoding = 4)	99.97 ±0.02	95.35 ±0.05

We evaluate seven variants of the CA-WNN model for MNIST classification, differentiated by their CA rulesets and the number of CA iterations. Mode 1 applies the Game of Life ruleset. Modes 2 and 3 are morphological CA with edge detection with 4-level and 8-level thermometer encoding, respectively. Mode 4 is morphological CA with Canny edge detection, with a Gaussian smoothing sigma of 1.0 and thresholds of 50 and 150 to capture weak and strong edges. These first four modes all employ two iterations of the CA rules. Mode 5 adopts the configuration of Mode 2 but performs only a single CA iteration, while Mode 6 follows the configuration of Mode 3 with just one iteration of the CA rules applied to the input images.

The baseline WiSARD model achieves high training accuracy but relatively lower test accuracy, indicating limited generalization. Incorporating CA significantly improves performance, with morphological CA-based models outperforming the Game of Life configuration. Among all variants, CA–WNN Mode 6 (single iteration with 8-level encoding) achieves the best overall performance, attaining near-perfect training accuracy and the highest test accuracy, demonstrating the effectiveness of combining fine-grained encoding with minimal CA iterations. Additionally, CA

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Mode 6 is evaluated by adding Gaussian noise to the MNIST dataset, resulting in negligible degradation in accuracy and demonstrating the robustness of the proposed model. Overall, the results highlight that carefully designed CA feature extraction enhances generalization while maintaining computational efficiency. Also, optimal feature extraction is achieved with minimal CA iterations, while additional iterations - beginning from the second - introduce morphological distortions, as shown in Fig. 5, that degrade model accuracy.

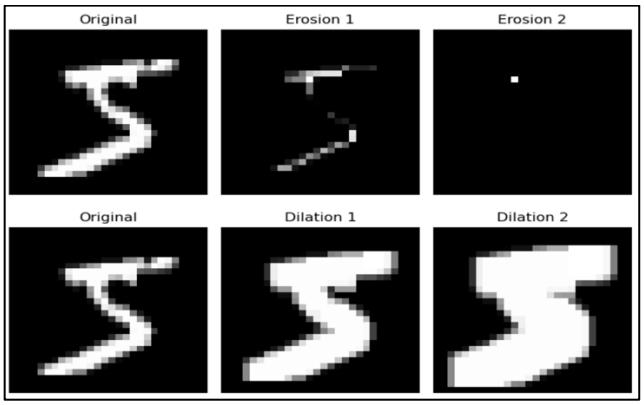


Fig. 5. Visualization of features obtained after the first and second iterations of CA for erosion and dilation.

Training and inference times were measured across different configurations of the CA–WNN model and evaluated on an NVIDIA RTX A6000 GPU with 47 GB of VRAM. CA–WNN Mode 2 achieved a training time of 1 min 9 s and an inference time of 1 min 51 s, whereas Mode 3 required 2 min 22 s for training and 3 min 45 s for inference, indicating higher computational complexity for the latter configuration due to increased Thermometer level encoding. The CPU-based Tiny CNN achieves a training time of 27.04 seconds and an inference time of 1.17 seconds with 98% test accuracy, demonstrating fast and efficient performance. Despite being slower, the CA–WNN offers a training-free, memory-based, and biologically inspired alternative, making it suitable for scenarios where gradient-based training is impractical or edge deployment is desired.

TABLE II. PERFORMANCE COMPARISON OF CLASSIFICATION MODELS ON CA-EXTRACTED FEATURES

Level encoding	Iteration	WiSARD	SVM	k-NN (k=3)
4	1	95.09±0.06	93.48±0.00	96.40±0.01
	2	93.47±0.12	93.97±0.00	94.99±0.02
8	1	95.22±0.03	93.96±0.00	96.32±0.01
	2	94.06±0.06	94.05±0.00	95.05±0.02
12	1	95.40±0.08	94.12±0.00	96.29±0.01
	2	94.28±0.01	94.25±0.00	95.06±0.00

Table II compares the classification accuracy of various models using identical CA-extracted features to determine whether the performance gains stem from the CA extraction stage or the classifier. The consistently high accuracy across all three classifiers indicates that the CA-based preprocessing pipeline is the primary contributor to the discriminative quality of the features. While k-NN achieves the highest accuracy, its $O(N \cdot d)$ inference complexity limits edge deployment; in contrast, WiSARD attains competitive accuracy with $O(1)$ inference, making it suitable for edge deployment.

Table III presents the effect of varying thermometer levels and CA iterations. Increasing encoding levels improves performance slightly; however, additional CA iterations consistently degrade accuracy due to feature distortion. With 4 levels, each 28×28 image is transformed into four binary layers by applying thresholds of 51.0, 102.0, 153.0, and 204.0. With levels 8, thresholds are 28.3, 56.7, 85.0, 113.3, 141.7, 170.0, 198.3, 226.7. While higher encoding levels enhance feature richness, they require more memory, making 4 and 8 levels a practical balance between accuracy and computational efficiency.

TABLE III. HYPERPARAMETER SWEEP OF THERMOMETER LEVEL ENCODING AND ITERATIONS ON CA-WNN PIPELINE

Level Encoding	Iteration	Test Accuracy (Mean ± Std)
4	1	95.04±0.09
	2	93.55±0.17
8	1	95.36±0.04
	2	94.01±0.04
12	1	95.34±0.04
	2	94.27±0.06

Table IV shows the effect of varying the RAM address size on the CA–WNN classification performance, using five pseudo-random mapping seeds per configuration. Smaller address sizes result in a larger number of RAM nodes per discriminator, while larger addresses reduce the number of RAMs. As the address size increases from 8 to 32 bits, the mean accuracy improves significantly from 76.42% to 95.22%, with the standard deviation decreasing, indicating more stable classification. Further increasing the address to 64 bits yields only a marginal gain (95.38%), suggesting that an address size of 32 bits provides an optimal balance between accuracy, stability, and memory requirements.

TABLE IV. SENSITIVITY ANALYSIS OF WISARD WITH VARYING ADDRESS SIZES AND MAPPING ON CA-WNN MODE 6

Address Size	RAMs per Discriminator	Mean Accuracy	Standard Deviation
8	2352	76.42	0.37
16	1176	91.85	0.39
32	588	95.22	0.08
64	294	95.38	0.05

Table V presents an ablation study analyzing the contribution of each CA rule individually and in combination. Combinations of rules demonstrated complementary benefits. Erosion with dilation or with edge detection, balanced noise removal, region expansion, and structural boundary capture. Dilation combined with edge detection achieved near-perfect training and high-test accuracy. Integrating all three rules produced the most stable and discriminative binary patterns, with erosion complementing dilation's region expansion and edge detection's boundary emphasis, enhancing feature robustness and spatial consistency.

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TABLE V. RULE ABLATION STUDY FOR CA-WNN MODE 6

Rule	Accuracy (%)	
	Train	Test
Erosion	62.34	57.90
Dilation	99.67	91.80
Edge Detection	99.97	92.12
Erosion + dilation	93.13	85.33
Erosion + edge detection	93.50	84.10
Dilation + edge detection	100	93.84
Erosion + edge detection + dilation	100	95.32

V. CONCLUSION

This study presents a hybrid CA–WNN framework for image classification that is, to our knowledge, the first to combine deterministic, rule-based CA feature extraction with a gradient-free, RAM-based weightless neural network classifier into a fully optimization-free pipeline. We explored enhancements to the CA-WNN model, focusing on morphological operators such as erosion, dilation, and edge detection, with their combination achieving competitive performance without relying on matrix multiplications or gradient-based optimization. Notably, optimal performance is achieved with a single CA iteration; additional iterations introduce morphological distortions that degrade accuracy, suggesting that lightweight feature extraction is preferable to deep iterative CA dynamics in this setting. While WNNs do not yet match the highest accuracies of convolutional networks, they train rapidly and are well-suited for resource-constrained, real-time applications. These results highlight the potential of automata-driven feature extraction for interpretable, rule-based learning and open avenues for future classification methodologies.

Future work will explore adaptive CA rule selection, multi-step automata dynamics, and extension to more complex datasets such as Fashion-MNIST, CIFAR-10, and medical imaging. Additionally, FPGA-based implementations will be investigated to enable low-power hardware deployment.

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